

Advanced Number Theory

ANSWER KEY



Divisibility rules and prime factorization

- Find the prime factorization of 1260.
 $1260 = 2^2 \times 3^2 \times 5 \times 7$.
- Determine whether 143 is prime, and justify your answer.
 $143 = 11 \times 13$, so 143 is not prime.

GCF, LCM, and their relationship

- Find the GCF and LCM of 36 and 60.
 $36 = 2^2 \times 3^2$, $60 = 2^2 \times 3 \times 5$. GCF = $2^2 \times 3 = 12$. LCM = $2^2 \times 3^2 \times 5 = 180$.
- Given that two numbers have GCF 8 and LCM 96, and one of the numbers is 24, find the other number, using the identity GCF $\times$ LCM = product of the two numbers.
GCF $\times$ LCM = $8 \times 96 = 768$. Other number = $\frac{768}{24} = 32$.

Modular arithmetic

- Find the remainder when 47 is divided by 6, and express this using modular notation.
 $47 = 6(7) + 5$, so the remainder is 5. In modular notation: $47 \equiv 5 \pmod{6}$.
- Find the last digit of 7^{100} , using the pattern of the last digits of powers of 7 (which cycle every 4: 7, 9, 3, 1).
Since the cycle length is 4, find the remainder when 100 is divided by 4: $100 = 4(25) + 0$, so the remainder is 0, which corresponds to the 4th position in the cycle (7,9,3,1): the last digit is 1.

Sums and products of integer sets

- Find the sum of all integers from 1 to 100, using the formula $S = \frac{n(n+1)}{2}$.
 $S = \frac{100(101)}{2} = 5050$.
- Find the sum of all even integers from 2 to 200.
This is an arithmetic series with first term 2, last term 200, and $n = 100$ terms. Sum
 $= \frac{n(a_1 + a_n)}{2} = \frac{100(2 + 200)}{2} = \frac{100(202)}{2} = 10100$.

Number sets and properties

- 9 Classify $\sqrt{16}$ as rational or irrational, and explain your reasoning.
 $\sqrt{16} = 4$, a whole number, which is rational (it can be written as $\frac{4}{1}$). Even though it involves a square root symbol, the result simplifies to an integer.
- 10 Explain why the sum of a rational number and an irrational number is always irrational.
If the sum of a rational number r and an irrational number x were rational (call it s), then $x = s - r$ would be the difference of two rational numbers, which is always rational — contradicting the assumption that x is irrational. So the sum must be irrational.
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Sequences and series

- 11 Find the 15th term of the arithmetic sequence with first term 6 and common difference 4.
 $a_{15} = a_1 + (15 - 1)d = 6 + 14(4) = 6 + 56 = 62$.
- 12 Find the sum of the first 8 terms of the geometric sequence 2, 6, 18, 54, ...
First term $a_1 = 2$, ratio $r = 3$. Sum $= \frac{a_1(r^n - 1)}{r - 1} = \frac{2(3^8 - 1)}{2} = 3^8 - 1 = 6561 - 1 = 6560$.
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Absolute value and inequalities with integers

- 13 Find all integer solutions of $|x - 5| \leq 3$.
 $-3 \leq x - 5 \leq 3$, so $2 \leq x \leq 8$. Integer solutions: 2, 3, 4, 5, 6, 7, 8.
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Combining number theory concepts in a multi-part problem

- 14 This question has three parts. (a) Find the prime factorization of 84 and 126. (b) Use the prime factorizations to find the GCF and LCM. (c) Verify your answer using the identity $\text{GCF} \times \text{LCM} = \text{product of the two numbers}$.
(a) $84 = 2^2 \times 3 \times 7$. $126 = 2 \times 3^2 \times 7$. (b) GCF (using lowest powers of shared primes): $2 \times 3 \times 7 = 42$. LCM (using highest powers of all primes present): $2^2 \times 3^2 \times 7 = 252$. (c) Verify:
 $\text{GCF} \times \text{LCM} = 42 \times 252 = 10584$. Product of original numbers: $84 \times 126 = 10584$ ✓, confirming the identity holds.
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Perfect squares, perfect cubes, and their properties

- 15 Find the smallest positive integer n such that $200n$ is a perfect square.
 $200 = 2^3 \times 5^2$. For $200n$ to be a perfect square, every prime must appear to an even power. Currently 2^3 needs one more factor of 2 to become 2^4 , and 5^2 is already even. So $n = 2$, giving $200(2) = 400 = 20^2$.

- 16** Determine whether 128 is a perfect cube, and justify your answer using prime factorization.
 $128 = 2^7$. For a perfect cube, the exponent must be divisible by 3. Since 7 is not divisible by 3, 128 is not a perfect cube.
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Counting divisors of a number

- 17** Find the number of positive divisors of 360, using its prime factorization $360 = 2^3 \times 3^2 \times 5$.
The number of divisors is found by adding 1 to each exponent and multiplying:
 $(3 + 1)(2 + 1)(1 + 1) = 4 \times 3 \times 2 = 24$.
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Working with consecutive integers and integer constraints

- 18** Find three consecutive even integers whose sum is 96.
Let the integers be $n, n + 2, n + 4$: $n + (n + 2) + (n + 4) = 96$, so $3n + 6 = 96$, giving $3n = 90, n = 30$.
The integers are 30, 32, 34.
- 19** The product of two consecutive positive integers is 156. Find the integers.
Let the integers be n and $n + 1$: $n(n + 1) = 156$, so $n^2 + n - 156 = 0$. Factor: $(n + 13)(n - 12) = 0$, giving $n = 12$ (rejecting the negative solution). The integers are 12 and 13.
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Digit sum and place value problems

- 20** A two-digit number has digit sum 11. When the digits are reversed, the new number is 27 more than the original. Find the original number.
Let the original number be $10t + u$ where t is the tens digit and u is the units digit: $t + u = 11$. Reversed number: $10u + t$. Given: $(10u + t) - (10t + u) = 27$, so $9u - 9t = 27$, giving $u - t = 3$. Adding $t + u = 11$ and $u - t = 3$: $2u = 14$, so $u = 7$, then $t = 4$. Original number: 47.
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A multi-part modular arithmetic and remainder problem

- 21** This question has three parts. (a) Find the remainder when 2^{10} is divided by 7. (b) Use the pattern of remainders of powers of 2 divided by 7 (which cycle) to find the remainder when 2^{50} is divided by 7. (c) Explain the general method for finding remainders of large powers using cycles.
(a) $2^1 = 2, 2^2 = 4, 2^3 = 8 \equiv 1, 2^4 = 16 \equiv 2 \pmod{7}$ — the remainders cycle with period 3: 2, 4, 1, 2, 4, 1, 2^{10} : since $10 = 3(3) + 1$, the remainder matches the pattern's 1st position: remainder 2. (b) 2^{50} : since $50 = 3(16) + 2$, the remainder matches the pattern's 2nd position: remainder 4. (c) The general method: find the length of the repeating cycle of remainders for the base's powers, then find where the target exponent falls within that cycle (using division with remainder), and match it to the corresponding position in the cycle — this avoids computing the actual (enormous) power directly.